OL OpenLine · Project Report

AI phone surveys in Indian languages, run from the survey form teams already use.

Gnani hackathon · Gnani Prisma + Timbre on Pipecat
github.com/pradneshfernandez/OpenLine
Try it: `make try` → localhost:8000/demo

1 The problem

Much of India's policy data (health, rations, crop losses, welfare) is still collected by enumerators visiting households or call-centre agents reading a form over the phone. Phone surveys are cheaper than visits, but slow and labour-heavy:

- **Few completes per day.** J-PAL South Asia plans for 10–14 completed 10–15 minute phone surveys per surveyor-day; most of the day goes on unanswered numbers and call-backs.
- **Many attempts.** In one J-PAL health survey only 27% of interviews were completed on the first call; the average was 3 attempts.
- **A week before the first call:** hiring, a 7-day training, translation, and one supervisor per 3–4 surveyors.

Teams also have carefully tested questionnaires and daily quality checks. A chatbot that rewords questions, or ignores their forms, is not usable for real fieldwork.

2 How real surveys run: what we studied

WHO	MAIN SOFTWARE	HOW QUALITY IS CHECKED
Research labs (J-PAL, IPA, World Bank DIME)	SurveyCTO	Daily high-frequency checks in Stata (*ipacheck*), back-checks (*bcstats*), DataSure dashboard
Humanitarian (REACH, UNHCR, WFP)	KoboToolbox, ODK	Daily cleaning, cleaning and deletion logs, HighFrequencyChecks (R)
Statistics offices (with the World Bank)	Survey Solutions	Supervisor and HQ approve or reject each interview

Shared ground: forms are **XLSForm** spreadsheets; phone rounds follow a call protocol (case lists, attempt statuses, call-backs, quotas) with 10–15% back-checks and audio audits.

We tested against 7 public forms from ODK, KoboToolbox, IFRC and SurveyCTO (including its CATI starter kit). They showed what breaks when a form is spoken, not shown: long option lists, grids, images, double questions and enumerator notes like *"DO NOT READ THIS ALOUD"*. That became OpenLine's 14-rule phone check.

3 The solution

OpenLine takes the XLSForm a team already has and runs it as a phone interview. The form stays in charge; the AI only listens.

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Respondent (8 kHz phone audio) → Gnani Prisma speech-to-text → Smart Turn v3.2 + last-word check (is the answer finished?) → LLM, through 6 typed tools only → form runner (skip logic, ranges, corrections) → Gnani Timbre voice → respondent
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- **Questions are read word for word**, one Pipecat Flows node per question. The LLM can only turn speech into a candidate value; the form validates it, so the AI can't skip, reword or invent questions.
- **Real conversation:** mixed languages ("मैं, मेरी wife और 3 बच्चे" saved as 5), changing an earlier answer ("only 2, not 3"), "pata nahi" saved as don't-know, numbers read back when unclear, re-asks on silence.
- **Fits existing work:** phone check, campaigns with retries and call-backs, an approve/reject review queue, CSV and KoboToolbox export with the original variable names, a recording and second-pass transcript per answer, and simulated respondents for testing.

4 How we did the cost analysis

Question: what does one completed 10-minute phone interview cost in India? Global benchmarks didn't answer it: an IPA review of 27 studies in 18 countries puts phone interviews at $11.97 per complete, mostly outside India, and our first slide (using Ghana and four-country figures) overstated the gap, so we replaced it with India data.

Source: J-PAL South Asia's own phone-survey budget (May 2020), which lists daily rates, productivity and calling costs for Indian fieldwork. We rebuilt the cost per completed interview from its line items:

HUMAN PHONE TEAM, PER COMPLETED INTERVIEW	
Surveyor: ₹675/day ÷ 10–14 completes a day	**₹48–68**
Supervisor: ₹750/day, one per 4 surveyors	**₹13–19**
Calls: ~11.5 min × ₹1.20/min (Exotel) + 18% GST	**~₹16**
Running cost (large study)	**₹80–100**
Setup: training ₹50k, translation, software, SurveyCTO, data packs ≈ ₹1.06 lakh ÷ 400 completes (J-PAL's 500-household example)	**+ ₹265**
All-in, 500-household study	**₹340–370**
OpenLine: 10 min × ₹6/min all-in (speech, LLM, telephony), up to ×1.5 for refused, dropped and call-back minutes	**₹60–90 est.**

> **Result:** about the same as a human team at scale, and 4–6× cheaper for small or urgent studies, where hiring, training and translation dominate. OpenLine can start calling the day the form is ready, and every call is recorded and checkable.

Caveats: J-PAL rates are from 2020 and not inflation-adjusted; project staff salaries are excluded, so human costs are, if anything, understated. Gnani's pricing is quote-based, so the ₹6/min figure uses Ringg AI's published India voice-agent pricing (Oct 2026) as a proxy.

5 Why Gnani and Pipecat

- **Gnani Prisma** streams speech-to-text in 10 Indian languages and hears mixed English and Hindi in one sentence; **Timbre** speaks all 10.
- **Pipecat** handles the real-time pipeline, phone audio (Twilio/Exotel media streams) and interruptions; **Pipecat Flows** gives one node per question, so the form, not the LLM, controls the call.

6 What we have tested, and what we haven't

TESTED	NOT YET, OR KNOWN LIMITS
Full calls in English and Hindi over an 8 kHz phone line: range checks, corrections, Hinglish answers, second pass. 201 automated tests.	A full AI-driven call in the 8 other languages (speech and call opening work in all 10; their prompts are drafts awaiting a native speaker).
Speech-to-text and voice in all 10 languages. English and Tamil mixed: 15 of 16 transcribed exactly, offline.	Interruptions with a live human voice; Tanglish in a live call.
End of turn: Smart Turn caught all 16 finished answers but only ~1 in 4 unfinished ones, so we added a check on the transcript's last word.	Known limit: Hindi and Tamil mixed in one sentence (0 of 6). No field pilot with real respondents yet.

Next: a pilot with a survey team on one real round, native review of the 8 draft languages, and AI back-check calls on human interviews, the lowest-risk way for a team to start.

Sources: P. Gupta, J-PAL South Asia, "Budgeting for phone surveys during the COVID-19 outbreak", May 2020 · IPA, "Remote Surveying in a Pandemic: Research Synthesis" · Ringg AI India pricing, Oct 2026 (proxy) · Daily, "Smart Turn v3.2" release notes · Public forms: ODK, KoboToolbox, IFRC, SurveyCTO sample libraries · Field practice: J-PAL, IPA (ipacheck, DataSure), UNHCR HighFrequencyChecks, World Bank Survey Solutions documentation.